WEEK - 4 Order Management System

1. Orders & Order_Details Table Design

A. Orders table

Stores the main information about each customer order.

Column	Data type	Description
`OID`	INT	Primary key
`CID`	INT	Customer ID, foreign key
`ODate`	DATE	Order date
`Total`	DECIMAL(10,2)	Total order amount
`Status`	VARCHAR(20)	Pending, Shipped, Delivered, or Cancelled

B. Order_Details table

Stores the products included in each order.

Column	Data type	Description
`DID`	INT	Primary key

`OID`	INT	Order ID, foreign key
`PID`	INT	Product ID, foreign key
`Qty`	INT	Quantity ordered
`Price`	DECIMAL(10,2)	Price per product

2. Updated ER Diagram

CUSTOMER

PK: CID

Name, Email

1 ↓ places many

ORDERS

PK: OID

FK: CID

ODate, Total, Status

1 ↓ contains many

ORDER_DETAILS

PK: DID

FK: OID, FK: PID

Qty, Price

Many ↑ refer to 1

PRODUCT

PK: PID

PName, Price

3. Sample Customer, Order & Order_Details Data

```
-- Sample customers
INSERT INTO Customer VALUES
(3, 'Arun', 'arun@gmail.com'),
(4, 'Divya', 'divya@gmail.com');

-- Sample orders
INSERT INTO Orders VALUES
(2, 2, '2026-10-25', 1000, 'Delivered'),
(3, 3, '2026-10-26', 50500, 'Shipped'),
(4, 4, '2026-10-27', 1500, 'Pending');

-- Sample order details
INSERT INTO Order_Details VALUES
(3, 2, 102, 2, 500),
(4, 3, 101, 1, 50000),
(5, 3, 102, 1, 500),
(6, 4, 102, 3, 500);
```

OID	Customer	Products	Total
1	Jeffrin	1 Laptop + 2 Mouse	₹51,000
2	Priya	2 Mouse	₹1,000
3	Arun	1 Laptop + 1 Mouse	₹50,500
4	Divya	3 Mouse	₹1,500

4. Customer Order History & Purchase Analysis Report

A. Customer order history

Displays each customer's order ID, order date, product, quantity, price, and status.

```
SELECT c.Name, o.OID, o.ODate, p.PName,
      d.Qty, d.Price, o.Status
FROM Customer c
JOIN Orders o ON c.CID = o.CID
JOIN Order_Details d ON o.OID = d.OID
JOIN Product p ON d.PID = p.PID
ORDER BY c.Name, o.ODate;
```

B. Total spending by each customer

```
SELECT c.Name, SUM(o.Total) AS Total_Spending
FROM Customer c
JOIN Orders o ON c.CID = o.CID
WHERE o.Status <> 'Cancelled'
GROUP BY c.CID, c.Name;
```

C. Product purchase analysis

Shows the total quantity sold for each product.

```
SELECT p.PName,
      SUM(d.Qty) AS Total_Sold,
      SUM(d.Qty * d.Price) AS Sales_Amount
FROM Product p
JOIN Order_Details d ON p.PID = d.PID
JOIN Orders o ON d.OID = o.OID
WHERE o.Status <> 'Cancelled'
GROUP BY p.PID, p.PName;
```

D. Average order value

```
SELECT AVG(Total) AS Average_Order_Value
```

```
FROM Orders
WHERE Status <> 'Cancelled';
```

E. Total sales amount

```
SELECT SUM(Total) AS Total_Sales
FROM Orders
WHERE Status <> 'Cancelled';
```

F. Order status report

```
SELECT Status, COUNT(*) AS Number_Of_Orders
FROM Orders
GROUP BY Status;
```

Expected purchase analysis

Based on the four sample orders above, assuming no cancelled orders:

Analysis	Expected result
Total orders	4
Total sales	₹1,04,000
Average order value	₹26,000
Laptops sold	2
Mice sold	8

Highest-spending customer	Jeffrin — ₹51,000

These totals assume Order 1 remains at ₹51,000 and no later updates change its total.